

KANNAN APPIYA SANTHARAM

Senior Lead Software Engineer

Chennai, India | Notice: 60 Days | Target: Dubai, UAE (Relocation Ready)

Phone / BOTIM / WhatsApp: +91 97902 47499

Email: as.kannan4@gmail.com

LinkedIn: linkedin.com/in/askannan

GitHub: github.com/kannan-santharam

Portfolio: kannan-ai-dev.vercel.app

PROFESSIONAL SUMMARY

Senior Lead Software Engineer with 10.5+ years of experience delivering high-performance web applications, monorepos, and scalable client platforms for global SaaS products. Specialized in frontend infrastructure, AI-assisted engineering workflows, LLM orchestration, custom Claude Code skills, Model Context Protocol (MCP) servers, and HTTP Streamable Web Streams (fetch + ReadableStream). Proven record of leading solo platform migrations, cutting build compilation times by 96%, and setting engineering standards across distributed engineering teams. Relocation ready for Lead Software Engineer or Engineering Manager roles in Dubai and the UAE.

CORE TECHNICAL HIGHLIGHTS & METRICS

- **96 Percent Build Speed Acceleration:** Solo executed a 3-week build system migration from Webpack 5 to Rust-powered Rspack across 12 packages at SuperOps, reducing cold-start compilation from 2 minutes down to 5 seconds and achieving HMR reloads under 50ms.
- **AI Test Authoring Platform & Real-Time Streaming:** Architected an automated end-to-end testing platform from scratch using React and Node.js, orchestrating LLM agents to auto-generate, execute, and self-heal Playwright test suites with real-time feedback streamed via HTTP Web Streams.
- **Production Claude Code Skills & MCP Servers:** Authored custom production Claude Code skills and engineered Model Context Protocol servers, establishing strict multi-tier write-scope guardrails to secure autonomous agent workflows across the codebase.
- **Dead Code Elimination (30k+ Lines):** Led codebase optimization using Knip static analysis, removing 30,000+ lines of unused dead code, unreferenced exports, and orphaned dependencies across 12 monorepo packages.
- **Enterprise Test Reliability:** Architected and maintained a comprehensive 232-spec Playwright regression test suite integrated into Jenkins CI/CD pipelines with zero tolerance for flaky tests.

WORK EXPERIENCE

SuperOps — Senior Lead Software Engineer

Jul 2022 to Present | Chennai, India

Role Progression: Senior Software Engineer → Lead Software Engineer → Senior Lead Software Engineer

- Architected an AI test authoring platform from scratch using React and Node.js, orchestrating LLM agents to auto-generate, execute, and self-heal Playwright test suites with real-time feedback delivered via HTTP Streamable Web Streams (fetch + ReadableStream).
- Authored custom production Claude Code skills and engineered Model Context Protocol (MCP) servers, establishing strict multi-tier write-scope guardrails to secure autonomous agent workflows across the codebase.
- Executed a solo Webpack 5 to Rust-powered Rspack migration in 3 weeks, reducing cold-start build compilation from 2 minutes to 5 seconds across 12 packages.
- Improved application load times through React lazy loading, route-level code splitting and per-package chunking strategy across the monorepo.
- Eliminated 30,000+ lines of dead code and unused dependencies across the monorepo using Knip static analysis.
- Maintained platform health with a 232-spec Playwright regression suite in Jenkins CI.
- Sole-authored a reusable API / UI SDK integration framework adopted across product teams, standardizing how features consume platform services.
- Built role-based access control (RBAC) gates and high-frequency polling dashboards using custom cached Apollo query hooks, balancing data freshness against network load.

Freshworks — Senior Software Engineer (Customer-Facing Engineering)

Jun 2018 to Jul 2022 | Chennai, India

Role Progression: Onboarding Engineer → Senior Software Engineer

- Worked directly with enterprise customers alongside Support, Customer Success and Sales teams to understand complex requirements, then partnered with Product Design to deliver UI solutions.
- Built UI web pages and REST API services across Freshworks products using React, JavaScript, Node.js, HTML and SASS.
- Built a Customer 360 dashboard consolidating all customer support, sales, and account information into a single unified operational view.
- Integrated tier 1 enterprise SaaS platforms including Jira, Salesforce, Zendesk and ServiceNow with Freshworks products through robust REST API services.
- Awarded Customer Champion Award for delivering critical customer UI features ahead of schedule.
- Created UI wireframes and design mockups in Balsamiq, mentored engineers on add-on development, and led code & UI design reviews.

- Infigenic, LLC — Software Developer

Jan 2018 to Jun 2018 | Bengaluru, India

 - Designed and implemented high-converting company web pages and product landing pages.
 - Built a seamless integration connecting Freshservice with DocuSign using REST API services; the product was subsequently acquired by Freshworks.
- Niche Video Media, LLC — Web Application Developer & Designer

Mar 2016 to Dec 2017 | Chennai, India

 - Designed and implemented responsive web pages, landing pages, and custom plugins for a commercial video-hosting platform.
 - Built a video player customisation feature from scratch with configurable player buttons, annotations, and call-to-action overlays.
 - Implemented an administrative dashboard that dynamically calculated Stripe pricing plans based on feature selection, storage tiers, and bandwidth limits.

TECHNICAL SKILLS MATRIX

Frontend Core: React, React Compiler, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, SASS

AI & Agentic Tech: Claude Code Skills, Model Context Protocol (MCP) Servers, Cursor IDE, LLM Orchestration, Write-Scope Safety, Amazon Bedrock AgentCore

State & APIs: GraphQL, Apollo Client, REST APIs, Zustand, Custom Cached Query Hooks, HTTP Streamable Web Streams (fetch + ReadableStream)

Architecture & Build: Monorepos, Rust-powered Rspack, Webpack 5, Vite, pnpm, Micro-Frontends, Knip Static Analysis, Design Systems, UI SDKs

Testing & DevOps: Playwright E2E Suite (232 specs), Node.js, Jenkins CI/CD Pipelines, AWS (EC2, S3, Route 53)

Leadership: Technical Team Leadership, Code Reviews, Developer Mentorship, Technical Knowledge Sharing, Engineering Governance

EDUCATION

Bachelor of Engineering (B.E.), Computer Science | K.L.N. College of Information Technology (2011 to 2015)